

CASE STUDY: Heart Disease Prediction Using Machine Learning

1. Problem Statement

Heart disease is one of the major causes of death across the world. Early detection of heart disease can significantly reduce the risk of severe health complications and improve patient survival rates. Traditional diagnosis methods rely heavily on medical expertise and manual analysis, which can be time-consuming and prone to errors. With the availability of medical data, machine learning techniques can be used to assist healthcare professionals in predicting heart disease more accurately and efficiently.

However, many existing machine learning models suffer from limitations such as improper feature selection, poor handling of imbalanced datasets, and reliance on single classifiers. These challenges reduce the overall effectiveness of prediction systems.

2. Objectives of the Case Study

The main objectives of this case study are:

- To analyze heart disease data using machine learning techniques
- To perform data preprocessing for improving data quality
- To apply feature selection to identify important medical attributes
- To develop and evaluate machine learning models for heart disease prediction
- To compare traditional models with a proposed ensemble-based approach
- To provide insights and recommendations based on the analysis

3. Dataset Description

- The dataset used in this study is the Heart Disease Dataset, obtained from a publicly available and widely used source. The dataset contains medical attributes such as age, gender, blood pressure, cholesterol level, maximum heart rate, and other clinical parameters. The target variable indicates whether a patient has heart disease or not.
- The dataset includes both numerical and categorical features and may contain missing values and class imbalance, which require proper preprocessing before model development.

4.Data Preprocessing

Data preprocessing is a critical step to ensure accurate and reliable model performance. The following preprocessing steps were applied:

- **Handling Missing Values:**
Missing values in the dataset were identified and handled using appropriate techniques such as mean or mode imputation.
 - **Data Normalization:**
Numerical features were scaled to a standard range to ensure uniform contribution during model training.
 - **Encoding Categorical Variables:**
Categorical attributes were converted into numerical format using encoding techniques.
 - **Feature Selection:**
Correlation-based feature selection was applied to remove irrelevant and redundant features, improving model efficiency.
 - **Train-Test Split:**
The dataset was divided into training and testing sets to evaluate model performance.
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5. Model Selection and Development

To predict heart disease, multiple machine learning models were developed and evaluated:

- **Logistic Regression:** Used as a baseline model due to its simplicity and interpretability.
- **Random Forest:** Applied to improve prediction performance by combining multiple decision trees.
- **Proposed FS-Ensemble Model:** An ensemble approach combining Logistic Regression and Random Forest with feature selection.

The proposed model aims to enhance accuracy, stability, and robustness compared to individual classifiers.

6. Model Evaluation

The models were evaluated using standard performance metrics:

- Accuracy
- Precision
- Recall
- F1-Score

These metrics provide a comprehensive evaluation, especially for medical and imbalanced datasets. The proposed FS-Ensemble model achieved better performance across all metrics compared to traditional models.

7. Visualizations and Insights

Various visualizations were generated to support the findings:

- Accuracy comparison bar graph
- Confusion matrix of the proposed model
- Feature importance bar graph

These visualizations provide clear insights into model performance, classification behavior, and the contribution of individual features to heart disease prediction.

8. Results and Discussion

The experimental results demonstrate that the proposed FS-Ensemble model outperforms Logistic Regression and Random Forest in terms of accuracy and reliability. Feature selection played a significant role in improving model performance by focusing on the most relevant medical attributes. The ensemble approach further enhanced prediction stability and reduced misclassification.

10. Conclusion

This case study demonstrates the effectiveness of machine learning techniques in predicting heart disease. The proposed feature selection-based ensemble model shows improved performance compared to traditional methods. The results highlight the importance of data preprocessing, feature selection, and ensemble learning in developing reliable medical prediction systems.

